

Elvis F. Elli

From: Agricultural & Environmental Letters <onbehalf@manuscriptcentral.com>
Sent: Thursday, July 2, 2026 9:24 AM
To: Elvis F. Elli; Maximo Nores
Cc: Maximo Nores; Elvis F. Elli; clsantos@iastate.edu; John Carlin; Jason Kelley; Anna.Locke@usda.gov
Subject: Agricultural & Environmental Letters - Revision Request for Manuscript AEL-2026-04-0060-RL

02-Jul-2026

Dear Dr. Elvis Elli,

Thank you for submitting your manuscript, "Night-time temperatures drive maize yield variability in the US Mid-South" (AEL-2026-04-0060-RL), for publication in the Agricultural & Environmental Letters.

After careful consideration, the expert reviewers and I feel your manuscript has merit but does not yet meet the criteria for publication in Agricultural & Environmental Letters. Therefore, we will reconsider a new version of the manuscript, provided that the suggestions of the reviewers and myself are addressed. I strongly urge you to revise your manuscript in accordance with the suggestions provided below and submitting a revision. If you choose to revise the manuscript, you must address each revision request or concern with a written response. Please highlight the changes to your manuscript within the document by using the track changes mode in MS Word or by using bold or colored text.

I encourage you to submit a graphical abstract along with your revised manuscript. Graphical abstracts of accepted articles are published with the eTOC and used for promotion. See our author instructions (<https://nam11.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.agronomy.org%2Fpublications%2Fjournals%2Fauthor-resources%2Fael-instructions&data=05%7C02%7Ceelli%40uark.edu%7C7800950168c943b8ebb008ded8459357%7C79c742c4e61c4fa5be89a3cb566a80d1%7C0%7C0%7C639185990547903456%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIlwLjAuMDAwMCIiOiIjXW4zMilslkFOljoitWVpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=m%2F4HMO%2FMIP5CRQkOh79DIXPvdVpgMRcaPyc1SQD2oL8%3D&reserved=0>) for details.

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Your revised manuscript should be submitted no later than 31-Aug-2026. If you need an extension, please contact me as soon as possible. If it is not possible for you to submit your revision in a reasonable amount of time, we may have to consider your paper as a new submission. Once again, thank you for submitting your manuscript to Agricultural & Environmental Letters and I look forward to receiving your revision.

Sincerely,

Dr. Anna Locke
Associate Editor, Agricultural & Environmental Letters

Editor: Dr. Alan Franzluebbbers, Alan.Franzluebbbers@usda.gov, ajfranzl@ncsu.edu Technical Editor: Dr. Peter O'Brien, peter.obrien2@usda.gov

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Author

The research presented in this paper investigates the effect of air temperature on grain yield of irrigated maize in eastern Arkansas during pollination/flowering. The authors used yield data from a variety testing program in combination with local weather station and reanalysis data to develop an empirical statistical model relating grain yield to weather variables. After validation, the authors used the model to simulate the effect of elevated air temperature on grain yield. Results showed that daily minimum air temperature, here used as a surrogate for the average nighttime air temperature, had a negative effect on grain yield and model simulations suggest that grain yield decreases by 3.7% per degree C of temperature increase for the conditions of the study domain (i.e., eastern Arkansas). Overall, results presented by the authors reinforce the evidence (e.g., Lobell & Field, 2007; Hatfield & Prueger, 2015; Hatfield, 2016) that maize yields decline under elevated air temperatures.

The manuscript fits the scope of the journal, and the topic has merit (i.e., nighttime air temperature effects on grain yield). However, I believe the manuscript needs to be substantially improved. Key concerns are as follows:

- The authors state that their study is representative of the "US Mid-South" (L65), when all experimental sites are in eastern Arkansas. This is misleading. Depending on how it is defined, the "US Mid-South" can include a range of states and production environments, which can be substantially different than the actual study domain. This extrapolation is not warranted.

- While I do not dispute the claim in L52 that minimum air temperature is correlated to nighttime temperature, using minimum air temperature as a variable of interest seems to defeat the purpose of the study when the goal stated by the authors is to better understand the effect of nighttime air temperature on grain yield. It is well-known that

minimum air temperatures occur just before sunrise and often for a short period of time. Nighttime air temperatures are typically taken as the values when the sun is below the horizon or when solar irradiances and PPFD values are below 2 W m⁻² or 5 μ mol m⁻² s⁻¹, respectively. Using solar angles or irradiances as filters to select nighttime values prior to averaging gives the actual nighttime air temperatures experienced by the crop. Using such values for nighttime air temperature will likely result in different results than those presented by the authors.

- The manuscript could be further edited to increase clarity and readability. The authors use acronyms that are not previously defined in many sections of the text (e.g., AIC, RMSE, and $|\beta|$).

Specific comments are as follows:

- Bullet point 1 (L11): “Minimum temperatures during pollination...”
- Bullet point 2 (L12): “Model simulations indicate that...”
- Line 22: “... driving yield penalties” to be consistent with terminology used in the manuscript.
- Line 23: “Simulations indicate that for every...”
- I would suggest avoiding the term “critical period”. Pollination or flowering are more accurate terms.
- Line 65: See my previous comment geographical representativeness. I would recommend not using the term “US Mid-South” and replacing it with “eastern Arkansas”.
- L76 and elsewhere in the text: Define acronyms at first mention (e.g., MinT).
- L85: Please clarify that this is a reanalysis data product.
- L86-88: Are these hourly, daily, or monthly values? Please clarify.
- L168-169: Another reason for quantifying nighttime air temperature more accurately.
- Fig. S1 and S2: Is this based on climate normals or measured values during the study period?

Reviewer: 2

Comments to the Author

The paper demonstrates a statistical association between June–July minimum temperature (NT) and maize yield, but the evidence is not yet sufficient to conclude that NT is the dominant climatic driver of yield variation or that the estimated yield penalties can be interpreted causally. Per my evaluation, the paper is publishable in principle at AERL, but I think it requires substantial strengthening of interpretation and methodology. I encourage authors to convince audience some thing like: NT is more important than the broader environmental system that accompanies warm nights. I have a few major concerns as follows:

1. NT may be acting as a proxy for several correlated environmental variables. Authors said ‘night-time temperature was the most important variable driving yields’, which is overclaimed. The June-July window (might be good in this paper. Most of time this fixed window hurts the conclusion for sure) is associated with Tx (maximum temp), VPD, cloud cover, crop respiration, and weather patterns (regional). The mixed-models have to be tested under multiple models (including or excluding possible predictors). It would be better if additional analyses or discussion are to distinguish whether Tn (Tmin) is acting as a direct physiological driver or as a proxy for multiple correlated environmental factors.
2. JJ fixed window may not represent actual crop exposure. ‘100% of maize area reached the critical period between early June and late July.’ might be true in Arkansas. However authors have 798 hybrids, and RM 103 to 121, multiple sites as well as planting dates, all of these could make reproductive phases variable substantially. I am not sure if authors could improve these: based them on crop calendars rather than calendar months.
3. Authors should be award that sensitive analysis is statistical, not physiological, nor the causality.
4. In addition, Authors assumed only Tn changed. This is risky and not undebatable. Sensitivity here is model sensitivity not the crop-climate sensitivity.

5. During the 2010 to 2024, the commercial maize hybrids changed substantially. Can you include hybrids as random effects in your models. This might complicate your paper but it could be as SI figure. Authors' -3.7% per C is within the range in regional scales (vs. the whole season's warming). It is possible Tx is not warming at all in Arkansas (I think so). Authors might find below references helpful:

Zhao, H., Tack, J.B., Kluitenberg, G.J., Kirkham, M.B., Sassenrath, G.F., Zhang, L., Wan, N., Liu, Z., Zhao, J., Ashworth, A. and Gowda, P.H., 2025. Concurrent improvements in maize yield and drought resistance through breeding advances in the US Corn Belt. *Nature Communications*, 16(1), p.9389.

Zhao, H., Yang, H., Avenson, T.J., Sassenrath, G.F., Kirkham, M.B., Welch, S.W., Zhang, L., Wan, N., Nelson, A.M., Gowda, P. and Lin, X., 2024. Nonlinear contributions of surface solar brightening to maize yield gains in the US Corn Belt. *Agricultural and Forest Meteorology*, 356, p.110169.

Jägermeyr, J., Müller, C., Ruane, A.C., Elliott, J., Balkovic, J., Castillo, O., Faye, B., Foster, I., Folberth, C., Franke, J.A. and Fuchs, K., 2021. Climate impacts on global agriculture emerge earlier in new generation of climate and crop models. *Nature food*, 2(11), pp.873-885.

6. 'an emerging threat to US maize production' should be cautious as authors tested under non-water stressed conditions. Similarly, 'The conclusions can be extrapolated to maize production systems in additional environments' is not supported (authors need evidence).

Associate Editor's Comments to Author:

Associate Editor: 1

Comments to the Author:

The manuscript was evaluated by two reviewers. Both note that the topic and dataset are of interest to AEL; however, both reviewers have concerns about the analysis methodologies. Please see reviewer comments for details. I also think for this analysis it would be useful to explicitly state the model(s) as an equation, particularly if the authors choose to address Reviewer 2's concerns.

With regard to Reviewer 1's concern about the use of Tmin instead of average dark-period NT, using Tmim is not an unprecedented strategy in high NT studies, and there are tradeoffs with using either parameter. The authors might consider the implications of choosing Tmin in the context of their specific objectives and provide strong justification for their final decision if they choose to submit a revised manuscript to AEL.

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